

sequence risk. In this way, even when the floor is reached, spending remains at about the same level as it was with the constant inflation-adjusted baseline strategy.

In this sense, the floor-and-ceiling strategy dominates the constant inflation-adjusted baseline strategy by providing the potential for higher spending throughout retirement. On the downside, it creates circumstances in which spending would be only slightly less than the baseline strategy. By allowing for higher spending, the floor-and-ceiling strategy ends up reducing the balance of remaining wealth after thirty years of retirement.

At the ninetieth percentile, spending is 20 percent higher than its initial level and 39 percent higher than the baseline level. At the median, spending is 15 percent below the initial level and 2 percent less than the baseline. The same is also true at the tenth percentile. This strategy allows for *some* variability in spending and spending reductions. The average spending reduction at the median is 1 percent, while it is 3 percent at the tenth percentile.

The next strategy in the exhibit is Vanguard's percentage floor-and-ceiling withdrawal strategy. This strategy is more variable in nature and has a bit more in common with the fixed-percentage withdrawal strategy than with the constant inflation-adjusted spending strategy. It's a floor-and-ceiling method that behaves more like a fixed-percentage strategy.

The initial spending level is 7.46 percent, and with this strategy, spending also tends to decrease by year thirty across the distribution of outcomes. It is 7 percent less at the ninetieth percentile, 64 percent less at the median, and 80 percent less at the tenth percentile. These are dramatic spending reductions, though they are not as dramatic as with a fixed-percentage spending strategy.

Relative to the baseline constant inflation-adjusted spending strategy, spending after thirty years is 65 percent higher at the ninetieth percentile, but it is 34 percent less at the median and 64 percent less the tenth percentile. Downside spending volatility is noticeable with a 1 percent